Step 7 : Decision

Since H_1 is a one-sided (right) alternative, elements of the critical region are determined by the rejection rule $z_0 > z_e$. Thus, it is a right-tailed test. For the given sample information, the observed value $z_0 = 3.87$ is greater than the critical value $z_e = 2.33$. Hence, the null hypothesis H_0 is rejected. Therefore, the mean breaking strength of the cables is significantly more than 1900 n/m^2.

Thus, the manufacturer's claim that the breaking strength of cables has increased by the new technique is found valid.

1.10 TEST OF HYPOTHESES FOR POPULATION MEAN (*POPULATION VARIANCE IS UNKNOWN*)

Procedure:

Step1 : Let μ and σ^2 be respectively the mean and the variance of the population under study, where σ^2 is unknown. If μ_0 is an admissible value of μ, then frame the null hypothesis as $H_0: \mu = \mu_0$ and choose the suitable alternative hypothesis from

(i) $H_1: \mu \neq \mu_0$ (ii) $H_1: \mu > \mu_0$ (iii) $H_1: \mu < \mu_0$

Step 2 : Let $(X_1, X_2, \ldots, X_n)$ be a random sample of n observations drawn from the population, where n is large ($n \geq 30$).

Step 3 : Specify the level of significance, α.

Step 4 : Consider the test statistic $Z = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}}$ under H_0, where $\bar{X}$ and S are the sample mean and sample standard deviation respectively. It may be noted that the above test statistic is obtained from Z considered in the test described in Section 1.9 by substituting S for σ.

The approximate sampling distribution of the test statistic under H_0 is the $N(0,1)$ distribution.

YOU WILL KNOW

It is important to note that the exact sampling distribution of Z is the *Student's 't'* distribution with $(n - 1)$ degrees of freedom, when n is small ($n < 30$). This hypotheses testing problem, when n is small, is discussed, in detail, in Chapter 2. When n is large, the *Student's* 't' distribution converges to the $N(0,1)$ distribution.

Step 5 : Calculate the value of Z for the given sample $(x_1, x_2, \ldots, x_n)$ as $z_0 = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$. Here, $\bar{x}$ and s are respectively the values of $\bar{X}$ and S calculated for the given sample.

Step 6 : Find the critical value, z_e, corresponding to α and H_1 from the following table

Alternative Hypothesis (H_1)	$\mu \neq \mu_0$	$\mu > \mu_0$	$\mu < \mu_0$
Critical Value (z_e)	$z_{\alpha/2}$	z_α	$-z_\alpha$

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